Fiscal Year:	FY 2018	**Task Last Updated:**	FY 04/10/2018
PI Name:	Rutkove, Seward M.D.		
Project Title:	Musculoskeletal Response to a Partial-gravity Analog in Rats: Structural, Functional, and Molecular Alterations		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Boston	**State:**	MA
Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Artificial Gravity. Appendix D: NASA HRP Artificial Gravity Opportunity
Start Date:	06/09/2016	**End Date:**	07/31/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2019 per NSSC information (Ed., 3/11/19) NOTE: Extended to 5/31/2019 per NSSC information (Ed., 10/23/18) NOTE: Extended to 9/30/2018 per F. Hernandez/ARC (Ed., 12/6/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bouxsein, Mary Ph.D. (Beth Israel Deaconess Medical Center/Harvard Medical Center)		
Grant/Contract No.:	NNX16AL36G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Health of the musculoskeletal system depends on exposure to mechanical loading. Removal or reduction of mechanical loading precipitates muscle atrophy and bone loss. Thus, despite rigorous exercise protocols and other countermeasures, muscle atrophy and bone loss are still virtually inevitable consequences of spaceflight. One approach for ameliorating the development of these untoward consequences of space travel is by the induction of artificial gravity during spaceflight via centrifugation. The strength of the artificial gravity depends both on the speed of rotation and diameter of the ship. While it is possible to achieve centrifugal force equivalent 1.0 G, it is also possible that lower levels of force may also be effective for preventing muscle atrophy and irreversible bone loss. In this study, we propose to evaluate the potential effectiveness of artificial gravity for the prevention of muscle atrophy and weakness and bone loss by studying rats using an innovative system for partial unloading that has been developed previously for mice by members of our team. We will study the musculoskeletal response of adult male rats (n=14 / group) following exposure to different degrees of unloading representing 0.2 G, 0.4 G, and 0.7 G as well as fully loaded animals for 1, 2, and 4 weeks. We will study the impact of this partial gravity analog environment on muscle and bone structure and function, including performing histological analyses, in situ force measurement of muscle contraction, as well as measurement of bone mass, microarchitecture, and strength. In addition, we will perform both serological and tissue molecular analyses to better understand the mechanisms and time course of development of these changes at different degrees of unloading. This work will also allow us to more effectively understand the relationship between the development of muscle atrophy and weakness and bone deterioration at varying levels of artificial gravity. At the conclusion of this study, we will have defined the potential impact of partial artificial gravity for prevention of musculoskeletal dysfunction and deterioration. This work will inform the future application of artificial gravity in human research and space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We believe that this research will positively impact in life on Earth since it will inform us about the effects of reduced muscle and bone use that occur normally in our daily lives and as a consequence of disease or injury. For example, using a cane may reduce the normal weightbearing of a single limb. While this obviously may be necessary from an assistive standpoint, this could lead to additional atrophy of the limb muscles and osteopenia of the bone.
Task Progress:	We have been making excellent progress on all 3 aims of the proposal. Over the past year, we have steadily studied multiple groups of rats exposed to different levels of partial weightbearing for periods of time from 1 to 4 weeks. Assessments have included electrophysiological and functional studies of muscle as well pQCT and microCT of hind limb muscles. We are in the process of completing extensive laboratory investigations as well. We effectively resolved a number of our early challenges to ensure that the animals tolerated the harness system well and were not able to disconnect themselves during the unloading. Our data shows that the analog is effective, inducing a dose-dependency to the degree of unloading; animals at progressively lower relative gravitational loads demonstrate decreasing muscle weight, reduced force generation with electrical stimulation and reduced grip strength, as well as lower cortical bone density. We anticipate completing most of the remaining animal data collection within the next 3-4 weeks, at which point our focus will turn mainly to tissue processing and molecular analyses.
Bibliography Type:	Description: (Last Updated: 05/21/2026)
Abstracts for Journals and Proceedings	Mortreux M, Ko F, Bouxsein M, Rutkove S. "A New Partial Gravity Analog Model in Rats to Investigate Musculoskeletal Alterations During Spaceflight." Presented at the 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Mortreux M, Ko F, Bouxsein M, Rutkove S. "A New Partial Gravity Analog Model in Rats to Investigate Musculoskeletal Alterations During Spaceflight." Presented at 2018 Orthopedic Research Society, New Orleans, Louisiana, March 10-13, 2018. 2018 Orthopedic Research Society, New Orleans, Louisiana, March 10-13, 2018. https://www.ors.org/Transactions/64/0689.pdf , Mar-2018
Articles in Peer-reviewed Journals	Mortreux M, Nagy JA, Ko FC, Bouxsein ML, Rutkove SB. "A novel partial gravity ground-based analogue for rats via quadrupedal unloading." J Appl Physiol (1985). 2018 Jul 1;125(1):175-82. Epub 2018 Mar 22. https://doi.org/10.1152/japplphysiol.01083.2017 ; PubMed PMID: 29565773 , Jul-2018
Articles in Peer-reviewed Journals	Ko FC, Li J, Brooks DJ, Rutkove SB, Bouxsein ML. "Structural and functional properties of bone are compromised in amyotrophic lateral sclerosis mice." Amyotroph Lateral Scler Frontotemporal Degener. 2018 Aug;19(5-6):457-62. Epub 2018 Mar 23. https://doi.org/10.1080/21678421.2018.1452946 ; PubMed PMID: 29569488; PubMed Central PMCID: PMC6141014 , Aug-2018